Hadi Mirza

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Education

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Electrical Engineering — *Expected June 2027*

Relevant Coursework: High-Speed Digital Design, Advanced Analog ICs, Analog Electronics + Lab, Signals & Systems + Lab, Logic Design + Lab, Computer Systems & C Programming

Technical Skills

PCB Design: KiCad, Altium; 2/4/6-layer boards, length-matched & impedance-controlled routing, DRC, Gerbers/BOM
Circuits: Mixed-signal sensor front-ends, MOSFET drive, LDO power trees, ESD protection, Li-ion charging
Simulation & Lab: LTspice, PSpice, MATLAB; oscilloscopes, SMD/THT soldering, board bring-up & debug
Embedded & Digital: C, C++, Python, Verilog; ESP32, Arduino, Xilinx Artix-7 (Vivado); I²C, SPI, UART; Git

Experience

Electrical Team Member – Low Voltage & Power Electronics — Oct. 2024 – Present
Formula Slug (FSAE Electric), UC Santa Cruz — *Santa Cruz, CA*

- Leading **KiCad** design of the FS5 **Load Slammer**, a **50 A transient-load tester** for characterizing the team's custom LV power supplies; moved from a hard-switched gate driver to a **closed-loop linear op-amp control** (OPA994 driving a 100 V / 1.7 mΩ MOSFET) for adjustable step size and slew rate
- Designed the measurement path with a 4-terminal 5 mΩ **Kelvin sense resistor**, **net-tied power/signal grounds**, and 50 Ω-terminated coax I/O for clean oscilloscope captures
- Contributed to **BMS and shutdown-circuit** schematics and wiring in KiCad supporting safe emergency shutdown and FSAE rules compliance, following team schematic standards and design reviews

Projects

6-Layer FPGA Development Board | *High-Speed Routing, Impedance Control* — Summer 2026

- Routed a custom **6-layer** FPGA board with I/O modeled after the Basys 3, using **dedicated power and ground planes** to minimize crosstalk and EMI
- **Length-matched differential pairs** and tuned **trace impedances** for 50 MHz signals; ran full DRC and schematic validation for manufacturability

Ground Bounce EMI Detector | *Altium Designer, Signal Integrity* — Summer 2026

- Designed a 2-layer **Altium** PCB with SMA test points to provoke and measure **ground bounce** from simultaneously switching logic
- Hand-soldered all SMD parts and captured **±20 mV** ground ringing on a 5 V, 1 MHz edge with a 500 MHz Keysight oscilloscope

Autonomous Gas-Mapping Robot (Senior Capstone, industry-sponsored) | *KiCad, Mixed-Signal* — Fall 2026 – Present

- Part of a team building a robot that autonomously surveys an area and produces a **gas-concentration map**; contributing to the **sensor board hardware** in KiCad
- Working on **MCU selection**, **power budgeting**, and the **low-noise analog front-end** for the gas sensor, with SPICE checks before layout

POBO Silent Haptic Pillow Alarm | *KiCad, ESP32, Li-Ion Charging* — Fall 2026

- Designed an **ESP32** carrier PCB driving 10 haptic motors through a logic-level **MOSFET low-side switch** with flyback protection, plus **USB-C Li-ion charging** (TP4056); iterated **5 revisions** through JLCPCB

Whiteboard CNC Drawing Robot | *C/C++, Arduino, Stepper Control* — 2026 – Present

- Built a 2-axis X-Y plotter on linear rails with a **dual-pen carriage** and **C/C++** firmware driving **stepper motors**; completed the first drawing test writing two-color text
- Designing a **custom PCB** to consolidate power delivery and control; developing image-to-toolpath conversion

Real-Time VGA Video Game | *Verilog, Xilinx Artix-7, Vivado* — Fall 2025

- Implemented a hardware-only game on an **Artix-7 FPGA** with a **VGA controller** timed for 640×480 @ 60 Hz and hierarchical **FSMs** in Verilog

Leadership

President, Multicultural Student Association, UC Santa Cruz — Lead 50+ members; organize events drawing 200+ attendees